Panskura Banamali College (Autonomous)

M.Sc. 4th Semester Examination - 2021 (Online Mode)

Subject: Computer Science **Paper: COS-403**

Paper Name: Elective-III (Combinatorial Algorithm)

F.M.: 40 **Time : 2 Hours**

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1. Answer any 02 questions **2×04=08**

a. What is difference between chaos and random?
b. What are the two main features of Genetic Algorithm?
c. What is hyper graph?

2. Answer any 02 questions **2×08=16**

a. Name and describe the main features of Genetic Algorithms (GA).
b. How the parameters affect the quality of solutions and the computational effort required by the Swarm Intelligence algorithms?
c. What are the types of optimization problems that can't be solved with meta-heuristic and heuristic algorithms?

3. Answer any 01 question **1×16=16**

a. Give an example of combinatorial problem. What is the most difficult in solving these problems?
b. Write short note of following:
   i. Optimization problem.
   ii. NP-hard problem
   iii. Pareto Optimal
   iv. Simulated annealing

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